Ashton E. Thomas

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New York, NY / New Jersey, USA

Education

University of Michigan - Rackham Graduate School 2025 – 2026
M.S.E. Computer Engineering, Computer Vision Concentration — *Ann Arbor, MI, USA*

summa cum laude; GPA – 4.00/4.00

Machine Learning, Robotic Kinematics & Dynamics, Applied Math for Robotics, Cryptography, Quantum Computing, Surveillance & Public Datasets, Statistics with R, Civil Engineering Directed Study

University of Michigan College of Engineering 08/2022 – 05/2025
B.S.E. Computer Science Engineering; Minor in International Engineering — *Ann Arbor, MI, USA*

summa cum laude; GPA – 3.8/4.0

Discrete Math, Data Structures & Algorithms, Computer Organization, Theoretical Computer Science, Extended Reality, Operating Systems, Web Systems, Probability, Data Analytics

Northwestern Michigan College – Dual Enrollment 2021 – 2022
A.S.A. Liberal Arts, summa cum laude, GPA – 3.96 / 4.00 — *Traverse City, MI, USA*

Traverse City West Senior High School 2018 – 2022
High School Diploma — *Traverse City, MI, USA*

Research Interests

Research interests include computer vision, artificial intelligence, machine learning, mapping algorithms, and reliable software systems, with emphasis on practical systems for geolocation, neural data analysis, and agentic engineering workflows.

Research Experience

Polk Lab — *University of Michigan* 09/2024 – 06/2025
Computational Neuroscience Research Assistant — *Ann Arbor, MI, USA*

- Processed fMRI data using FS-FAST and Makefiles to study age-related neural dedifferentiation and its effects on neurocognition in healthy aging and Alzheimer's disease.
- Contributed to machine learning workflows measuring how different languages affected neural distinctiveness in the human brain, using speech-versus-background-noise separation to support the analysis, and connecting computational models with human-like auditory processing.
- Built Python data-processing tools with Prof. Thad Polk for CSV datasets with 1,000+ columns, including wildcard column filtering, demographic parsing, node-hierarchy operations, and reproducible row filtration.
- Used MATLAB, Python, Git, and structured research workflows to support computational neuroscience analysis and data management.

Selected Projects

Geoguessr AI, Computer Vision 2023 – 2024

- Designed and implemented a modified ResNet-50 architecture for geographic location identification from images.
- Fine-tuned the model using 61,000 images, addressing challenges related to lighting and seasonal variations.
- Achieved approximately 90% accuracy in predicting U.S. geographic locations from visual data.

Google Search Engine, Web Systems 2024

- Engineered a scalable search engine leveraging a segmented inverted index implemented with MapReduce for efficient data processing.
- Integrated tf-idf for text analysis and PageRank for link analysis to improve the relevance of search results.
- Developed a REST API to deliver fast and accessible search query results.
- Designed and implemented a user-friendly interface for seamless interaction with the search engine.

Study Group Coordinator, Quantum Computing 2024

- Designed and developed a Study Group Scheduler with Quantum algorithms leveraging Grover's algorithm for efficient group formation under CNF constraints.
- Created and implemented Bitflip and Phase Oracles to translate CNF constraints into quantum operations.
- Engineered a quantum counting circuit to estimate the number of feasible solutions for optimal scheduling.

- Utilized quantum algorithms to optimize solution search, showcasing advanced problem-solving techniques in quantum computing.

Virtual Memory Pager, Operating Systems 2025

- Built a virtual-memory pager managing per-process arenas, page tables, physical frames, swap-backed pages, and shared file-backed mappings.
- Implemented page-fault handling, copy-on-write behavior, dirty/reference tracking, swap allocation, file-block sharing, eviction, and a clock-style page replacement policy.

Networked File Server, Operating Systems 2025

- Implemented a socket-based network file server supporting read, write, create, and delete operations over a block-based disk abstraction.
- Designed validation and concurrency logic for ownership, paths, users, permissions, capacity limits, shared/exclusive locks, threaded request handling, and safe cleanup.

Invariant Learning Experiments, Machine Learning 2025

- Compared empirical risk minimization, invariant risk minimization, and IRM-Games under distribution shift across controlled experimental environments.
- Evaluated model behavior on Colored MNIST, synthetic chain-equation data, and financial time-series settings with spurious correlations, trend regimes, and volatility shifts.

Professional Experience

Amazon Web Services 06/2025 – Present
Software Development Engineer I *New York, NY, USA*

- Build developer-insights platforms for software delivery health, observability, and engineering analytics across large-scale internal software systems as part of a 12+ person AWS team.
- Design and operate AWS data pipelines using Python, S3, Glue, Athena, CloudWatch, and infrastructure-as-code workflows for analytics and reporting.
- Develop privacy-aware data availability and access-control validation, including integration tests for row-level and column-level security across analytics datasets.
- Designed AI-assisted engineering automation that coordinates task intake, code generation, code review lifecycle management, reliability checks, and operational observability.
- Support production operations through canary checks, alarms, dashboarding, bug investigation, on-call readiness, and incident-driven reliability improvements.
- Managed Amazon company-wide employee data for 100,000s of employees as part of a 10-engineer team, including collection, processing, and executive-level inferences.

Amazon 09/2024 – 11/2024
Software Development Engineer Intern *Boston, MA, USA*

- Worked on Alexa endpoint experiences, designing, testing, and optimizing customer-facing device functionality with a team of more than 20 engineers.
- Implemented software features using React Native, Kotlin, TypeScript, and related technologies that shipped into Alexa+, applying production engineering and responsive design practices.

Ground Vehicle Systems Center 05/2024 – 07/2024
Software Development Intern *Warren, MI, USA*

- Leveraged MagicDraw & Excel to design databases for Jira tickets and hardware, leading to improved efficiency.
- Created Python scripts to parse large csvs with 1000s of datapoints to update integrated networks in Jira.

Madi Taylor Photo 06/2022 – 07/2024
Full Stack Developer *Traverse City, MI, USA*

- Developed and maintained the corporate website, crafting a cohesive user interface with HTML, CSS, and JS.
- Implemented robust back-end payment solutions and form validation to streamline user transactions.

Selected Papers

Geoguessr AI: A Look into AI Geolocation Accuracy 2024

Teaching & Mentoring

Peer Tutoring and Study Group Leadership University of Michigan

- Tutored peers in mathematics, computer science, and related technical coursework, supporting problem solving, conceptual review, and exam preparation.
- Led university computer science study groups ranging from approximately 4–20 students, coordinating review sessions around programming, algorithms, systems, and theory coursework.

Engineering Onboarding Amazon

- Onboarded 2 new graduate engineers into team codebases, operational practices, software delivery workflows, and engineering standards.

Technical Skills

Coding Languages: ARM, C/C++, CSS, HTML, Java, JavaScript, Kotlin, LaTeX, MATLAB, Python, SQL, TypeScript
Cloud & Data: AWS, Lambda, S3, Glue, Athena, Redshift, CloudWatch, CDK
AI/ML & Research: Computer vision, machine learning, neural data analysis, mapping algorithms, PyTorch, Keras, TensorFlow
Developer Tools: Git, VSCode, CI/CD pipelines
Engineering & Mapping Tools: AutoCAD, ArcGIS, Excel, Jira, MagicDraw
Frameworks: Flask, React, React Native
Libraries: Jinja, NumPy, Pandas, Qiskit
Operating Systems: Mac, Windows

Certifications

Engineering Across Cultures: University of Michigan College of Engineering — 02/2024
Understanding and Visualizing Data with Python: Coursera — 08/2023
Advanced Styling with Responsive Design: Coursera — 08/2023

Selected Scholarships & Honors

Scholarships: Mary Sue and Kenneth Coleman Student Scholarship; Michigan Regents Merit Scholarship; Riopelle/Dowden Technical Award; Robert Paul Dost Award; Ernest B. Isaacsen Award; Guy M. Wilson Award; Marilla Church of the Brethren Scholarship
Academic Honors: Dean's List, University of Michigan; Dean's List, Northwestern Michigan College

Languages

English: Native
German: B1
Spanish: A2

Leadership & Activities

Michigan Data Science Team — 2024
University of Michigan EV & Mobility Scholars: student program focused on electric vehicles, mobility systems, and transportation technology — 2023 – 2025
MHackers — 2022 – 2024
First Robotics — 2021 – 2022

Volunteering & Service

The Kosciuszko Foundation: volunteer with a Polish-American foundation focused on education, cultural exchange, and community programming — 10/2025 – Present
National Honor Society — 08/2021 – 05/2022

Interests/Hobbies

Artificial intelligence, backpacking, computer vision, geopolitics, hiking, machine learning, mapping algorithms, neuroscience, travel photography

References

Thad Polk
Professor of Psychology, University of Michigan; Principal Investigator, Polk Lab
Research supervisor for computational neuroscience work

Jeremy Muldavin
Mentor; Ph.D., University of Michigan; long-time researcher at MIT Lincoln Laboratory
Technical and research mentor